SeeWandel Climate

Interreg
Alpenrhein-Bodensee-Hochrhein

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Regime Shifts in Plankton Dynamics: What Drives Them and Can We Predict Them?

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www.seewandel.org

Ecomeets @ Eawag, Dübendorf
28 October 2026

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Slides, code and data shared openly
leomarameo7.github.io/science_communication.html

Acknowledgements

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Prof. D. Straile



Prof. U. Gaedke



Dr. J. Alexander



Dr. P. Teiber-Sießegger

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

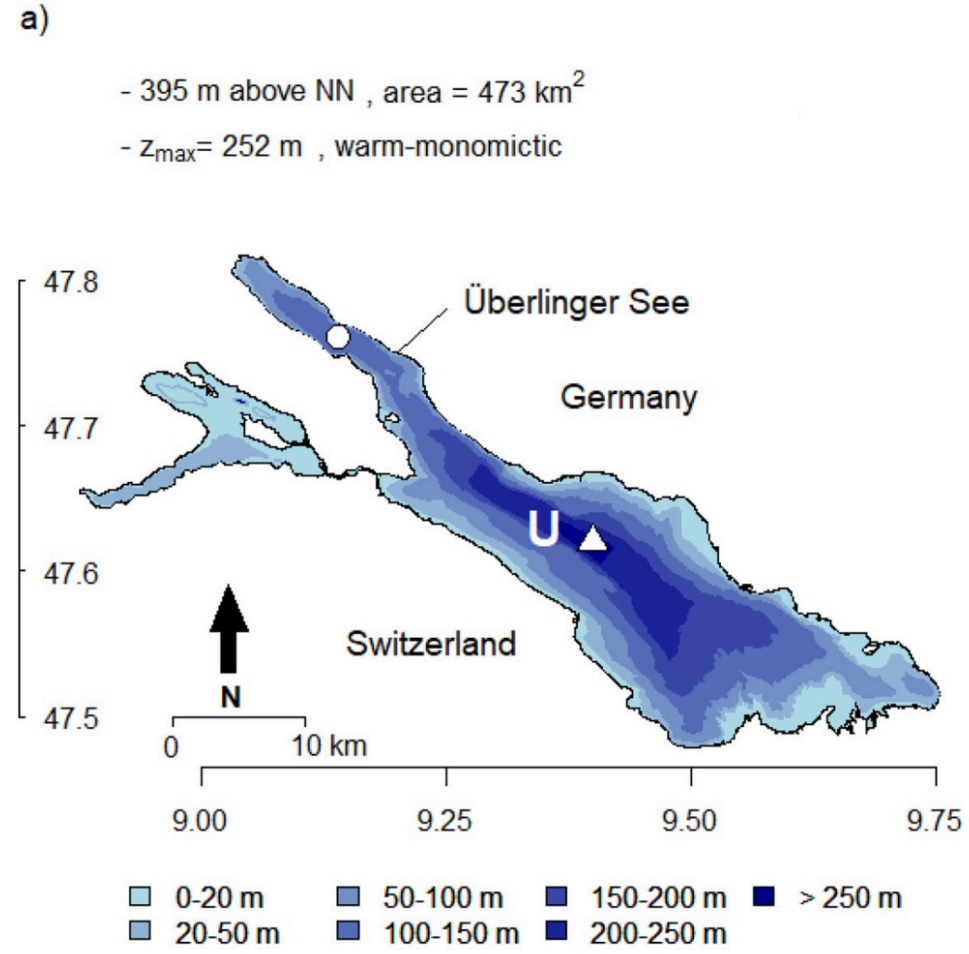
2. Whether they propagate and can be predicted is still open.

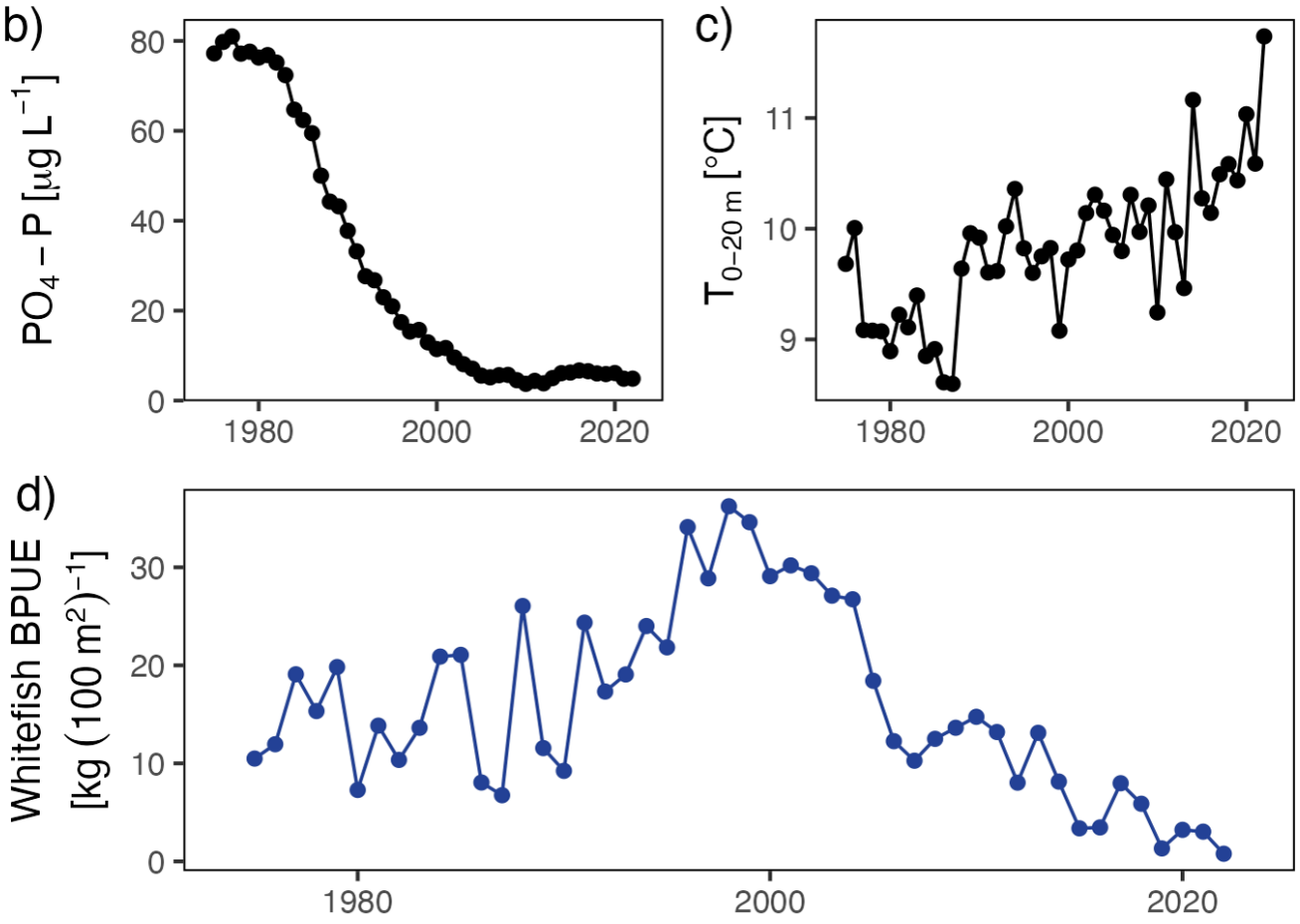


Outline

1. What we know: background
2. Research questions
3. Methods: our pipeline
4. Results: detection of regime shifts
5. Explain regime shifts
6. Propagation of regime shifts
7. Conclusions

What we know: background



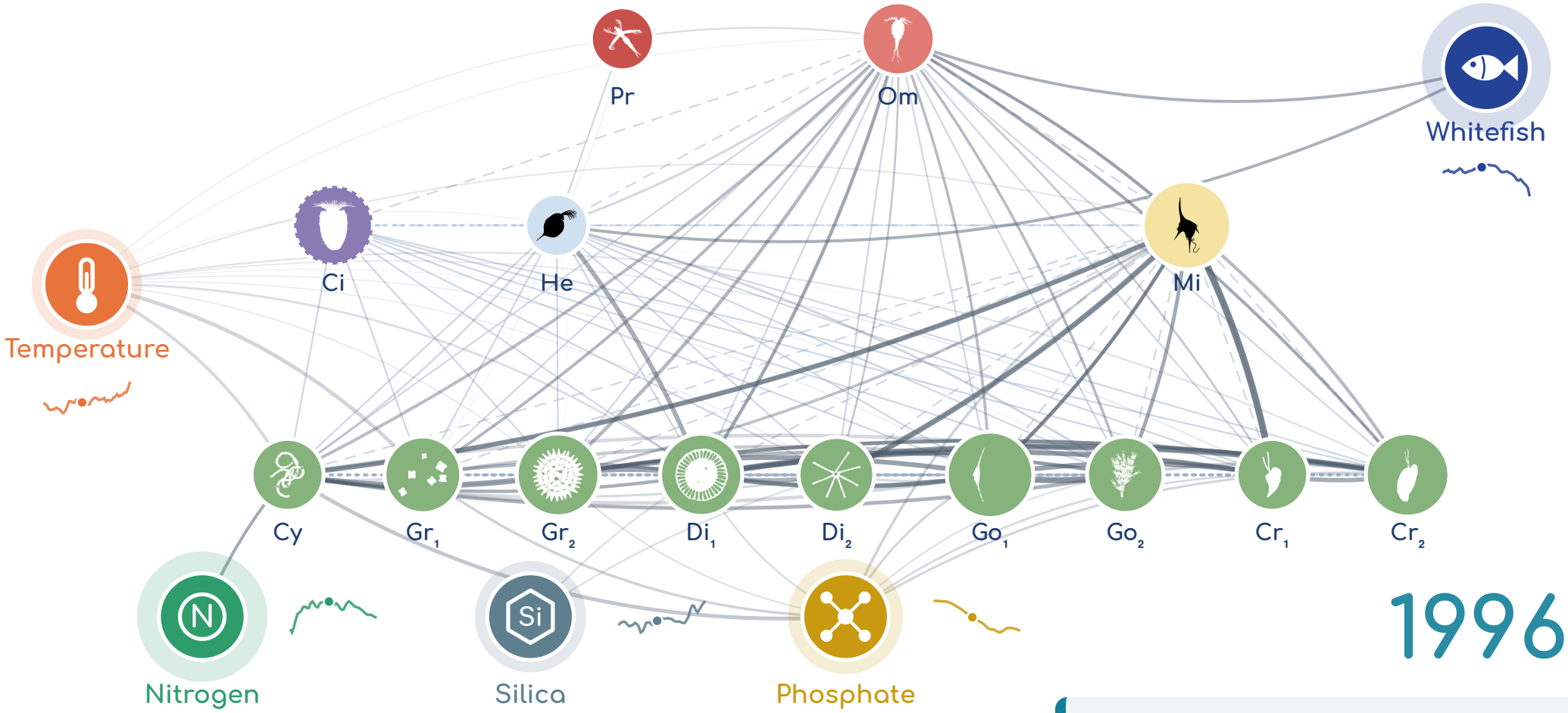


Lake Constance · 473 km²

Phosphate ↑ until 1980, then ↓

Temperature ↑ about 1 °C

What we know: the plankton community



Study period: **1975–2022** · monthly time series

The plankton community changes across time

Network modified from Merz et al. 2023, Nat. Clim. Change. Links without a strength: minimum intensity.

Research questions



1 · Detect

Abrupt shifts?



2 · Explain

What drives them?



3 · Propagate

Does a shift spread?



4 · Predict

Can we forecast them? (in progress)

Methods: our pipeline

1

Data

Monthly data 1975–2022

2

Detect

BEAST + two criteria

3

Explain

GP-EDM + hypotheses

4

Propagate

Shift-removal test

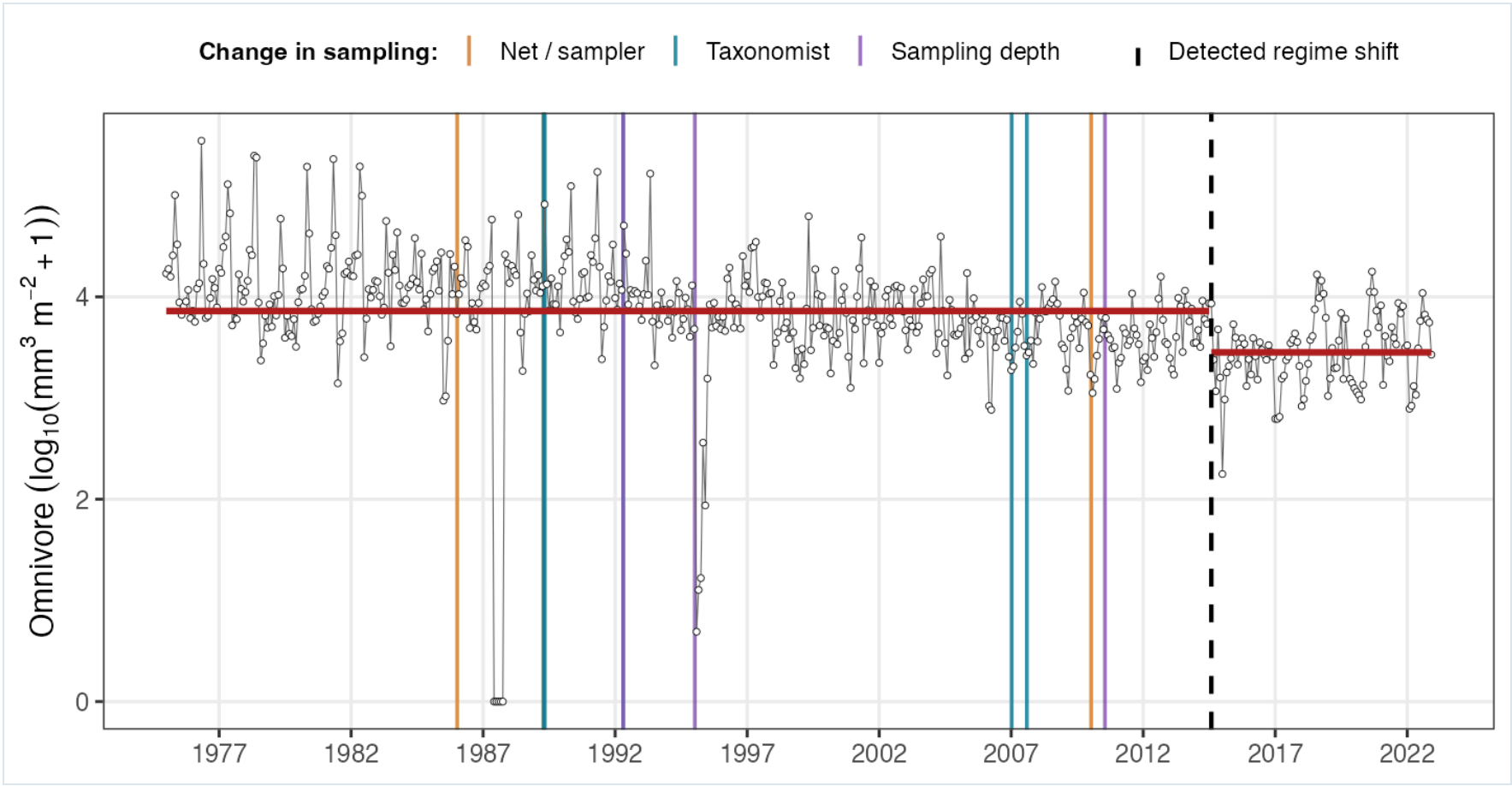
5

Predict

Forward validation

WORK IN PROGRESS

Results: detection of regime shifts





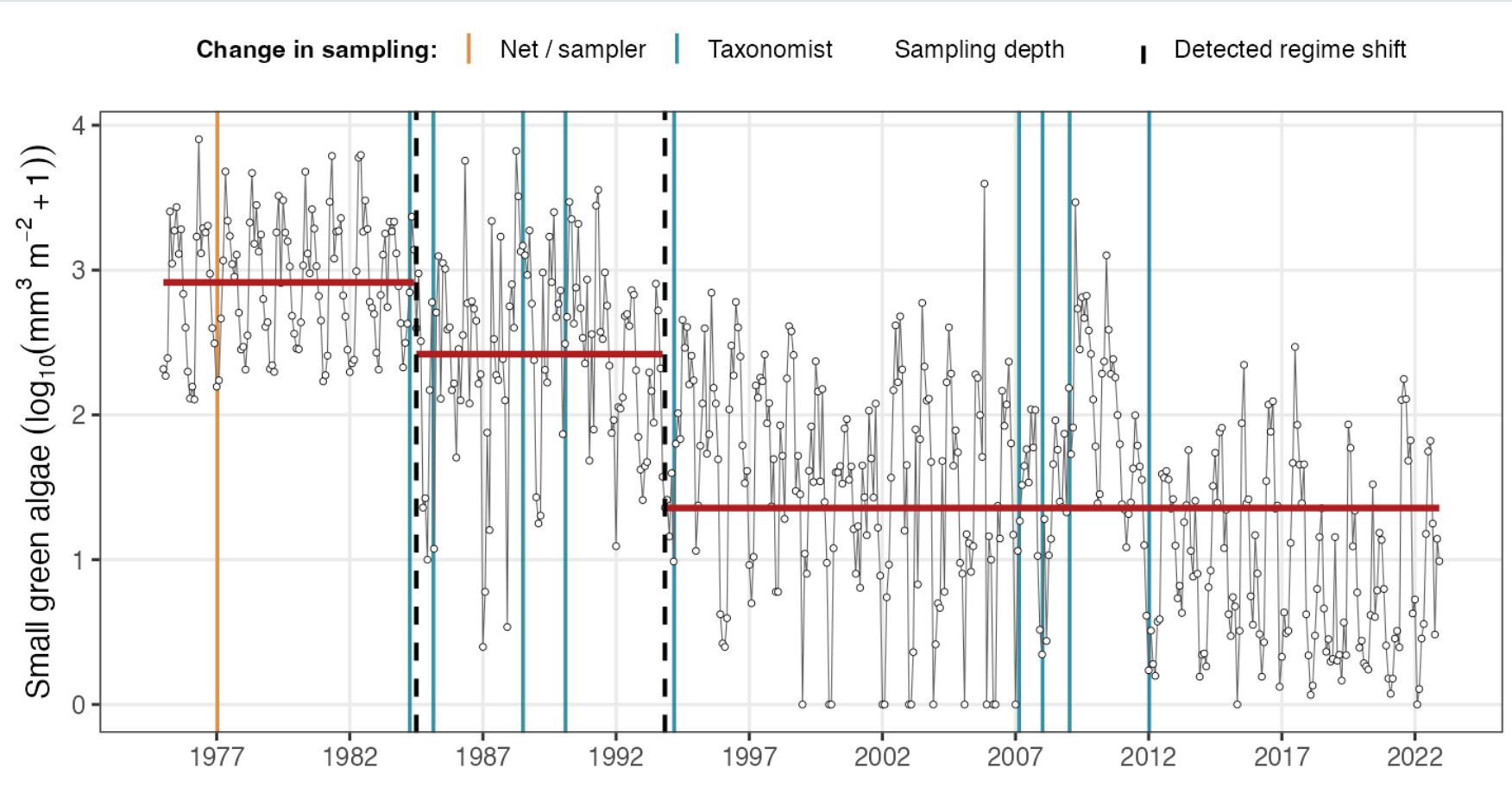
Shift: Aug 2014

Abruptness 1.19
jump size vs. usual swings (> 1 = abrupt)

Omnivores, monthly biovolume. Red: period means. The shift does not coincide with any change in sampling.

The omnivores dropped abruptly in 2014

Results: detection of regime shifts



Small green algae, monthly biovolume. Red: period means.

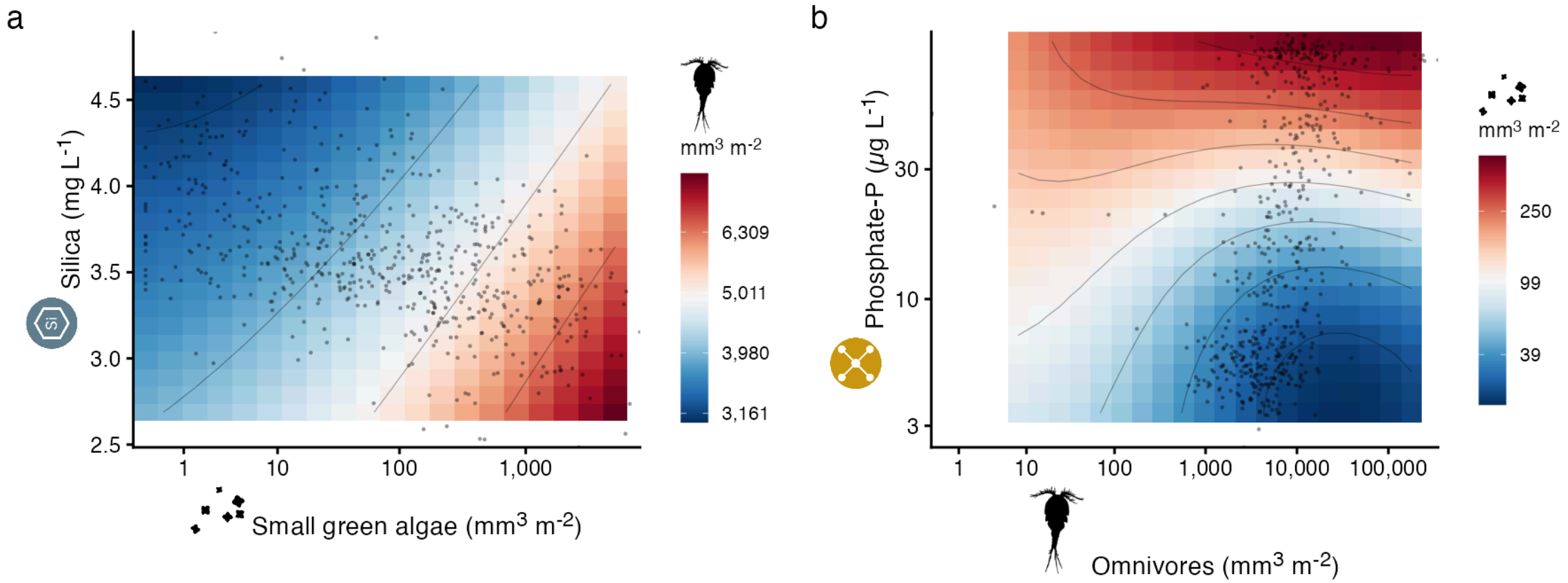
Jul 1984
abruptness 1.20

Nov 1993
abruptness 1.15

Both ≤ 4 months from a taxonomist change

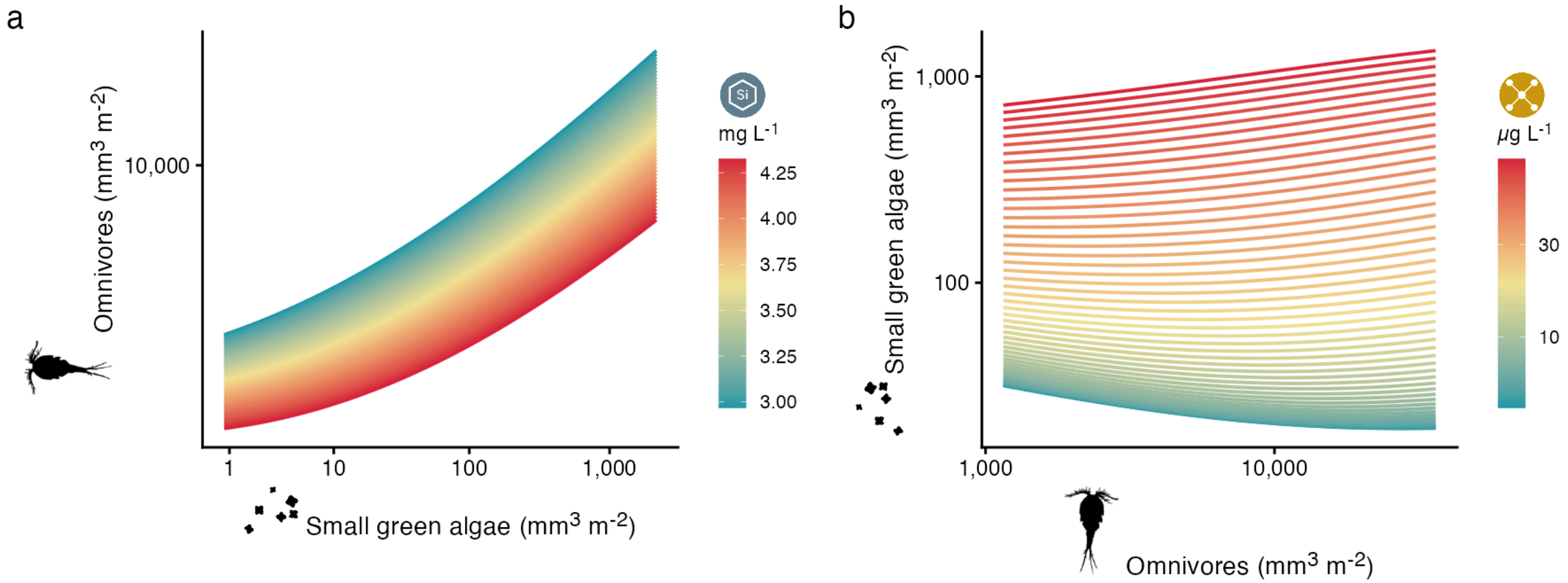
Small green algae shifted twice

Explain regime shifts



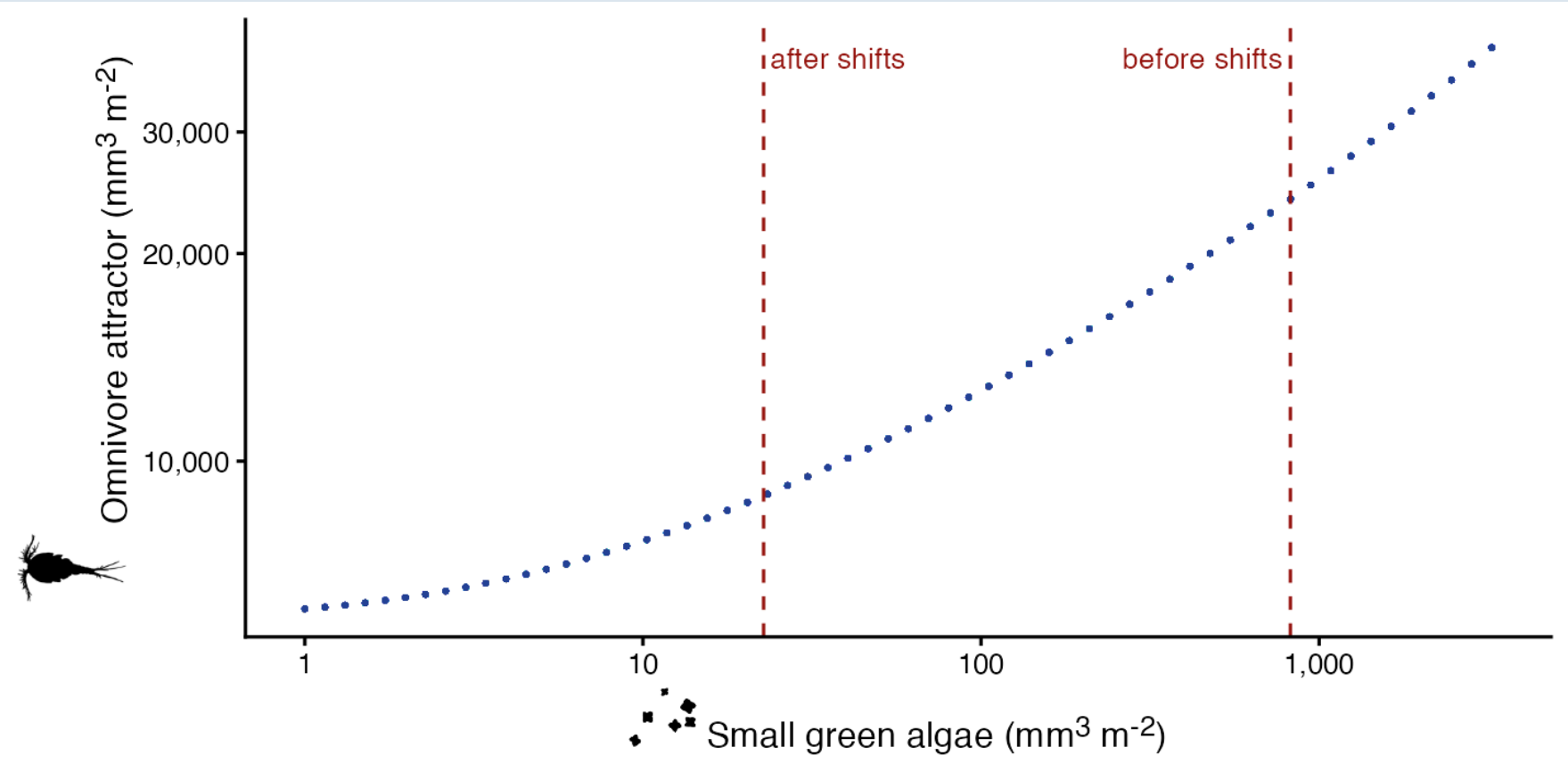
Phosphate drives small green algae

Explain regime shifts: partial dependence



Green algae favour omnivores

Regime shift: prediction



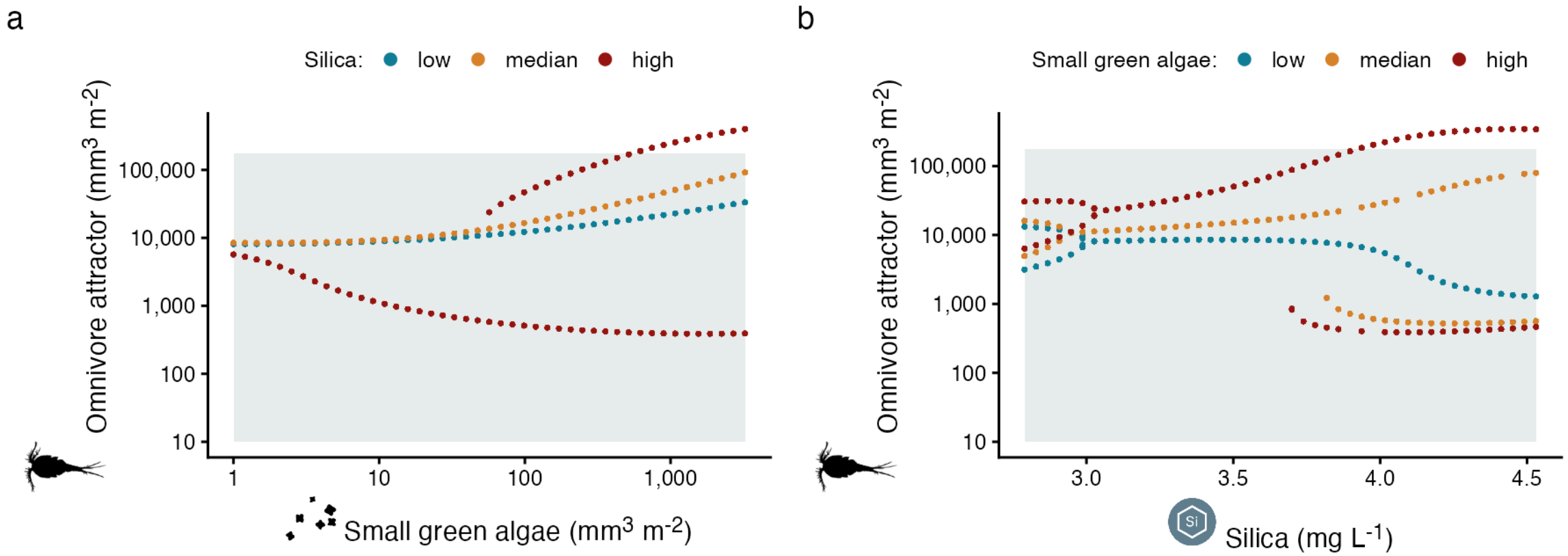
Model: omnivores from their own past, small green algae and the key abiotic drivers

LOO R^2 = **0.56**

Every start reaches **one** attractor

No bifurcation: smooth response

Regime shift: prediction (2 drivers)



Two stable states at high silica

Propagation of regime shifts

1 · Real predictor



2 candidate pairs

0 testable

Predictor shifted **after**

2 · Shift-free surrogate (× 100)



Gap between real and surrogate prediction = evidence for propagation

No evidence yet: too few shifts

Conclusions



Shifts are detectable



Biotic drivers matter



Propagation, prediction: open

Take home message

1. Plankton regime shifts can be detected and partly explained by biotic drivers.

2. Whether they propagate and can be predicted is still open.



Acknowledgements

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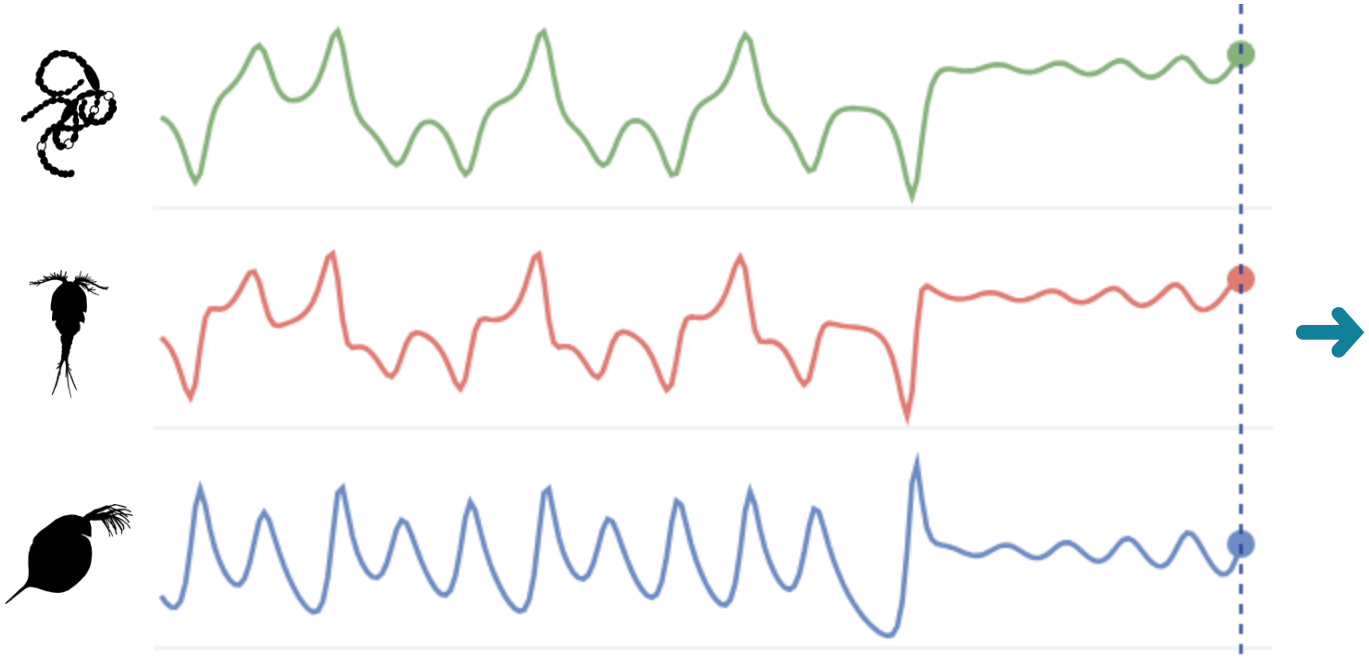


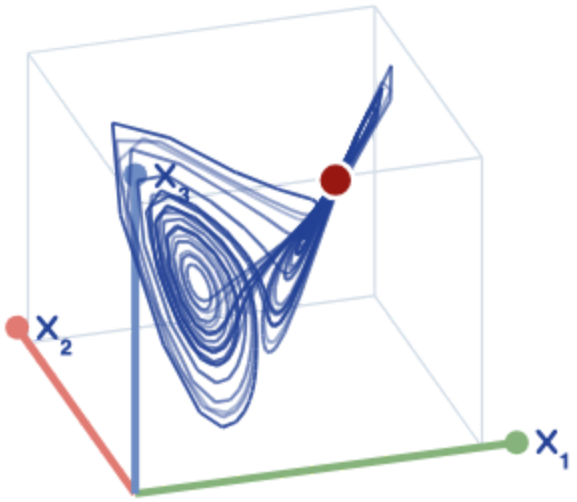
Dr. P. Teiber-Sießegger

Supplementary materials



What is GP-EDM?





State

$x_t = (x_{1,t}, x_{2,t}, x_{3,t})$

Dynamics (unknown)

$x_{t+1} = f(x_t) + \varepsilon_t$

GP learns f

$k(x,x') = \sigma^2 \exp(- \sum_i \phi_i (x_i - x'_i)^2)$

Main advantage of EDM

No model equations: the data reveal the dynamics

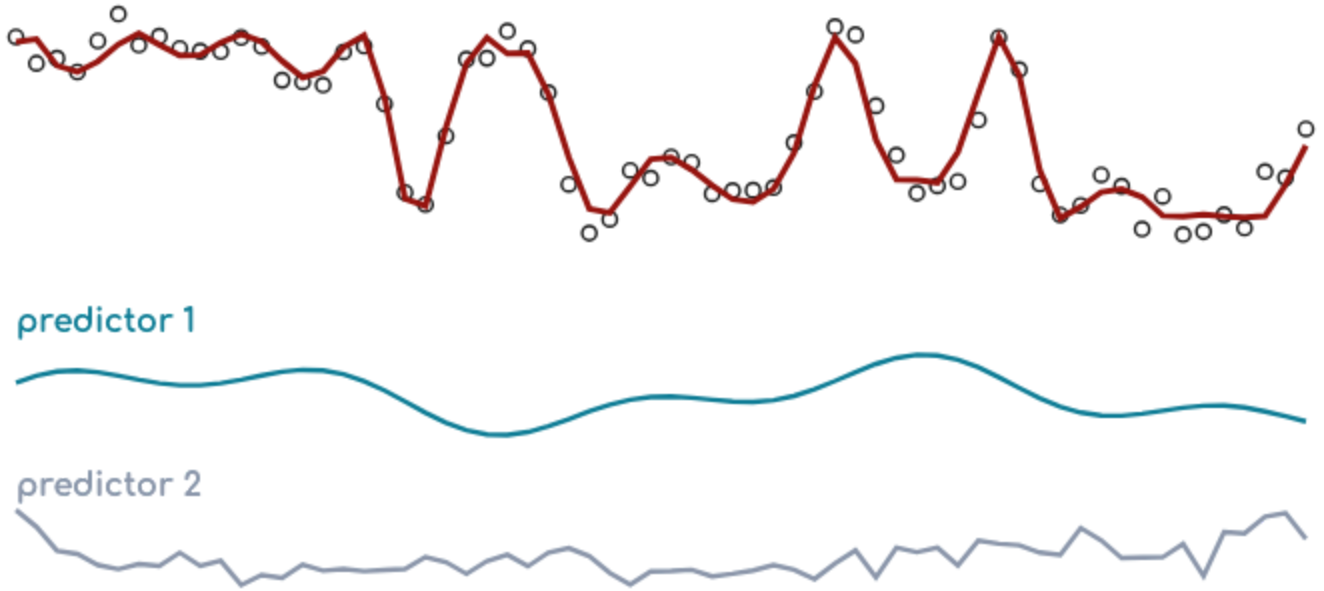
Takens' theorem (key assumption)

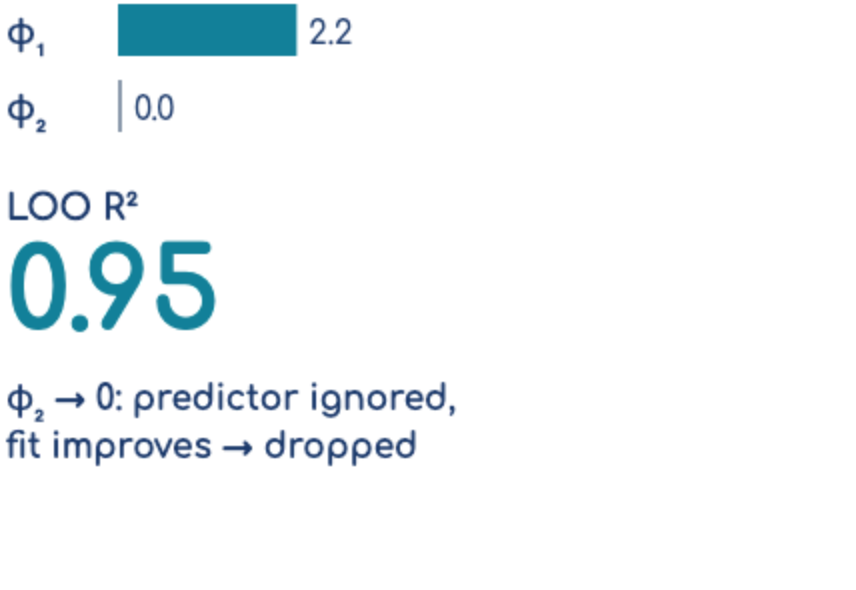
Delay coordinates rebuild the attractor

$E \geq 2d + 1$

Illustration: simulated system. Close states have close futures (Takens 1981; Sugihara & May 1990)

Automatic relevance determination (ARD)





One ϕ per predictor

$k(x,x') = \sigma^2 \exp(- \phi_1(x_1 - x'_1)^2 - \phi_2(x_2 - x'_2)^2)$

Why drop predictors?

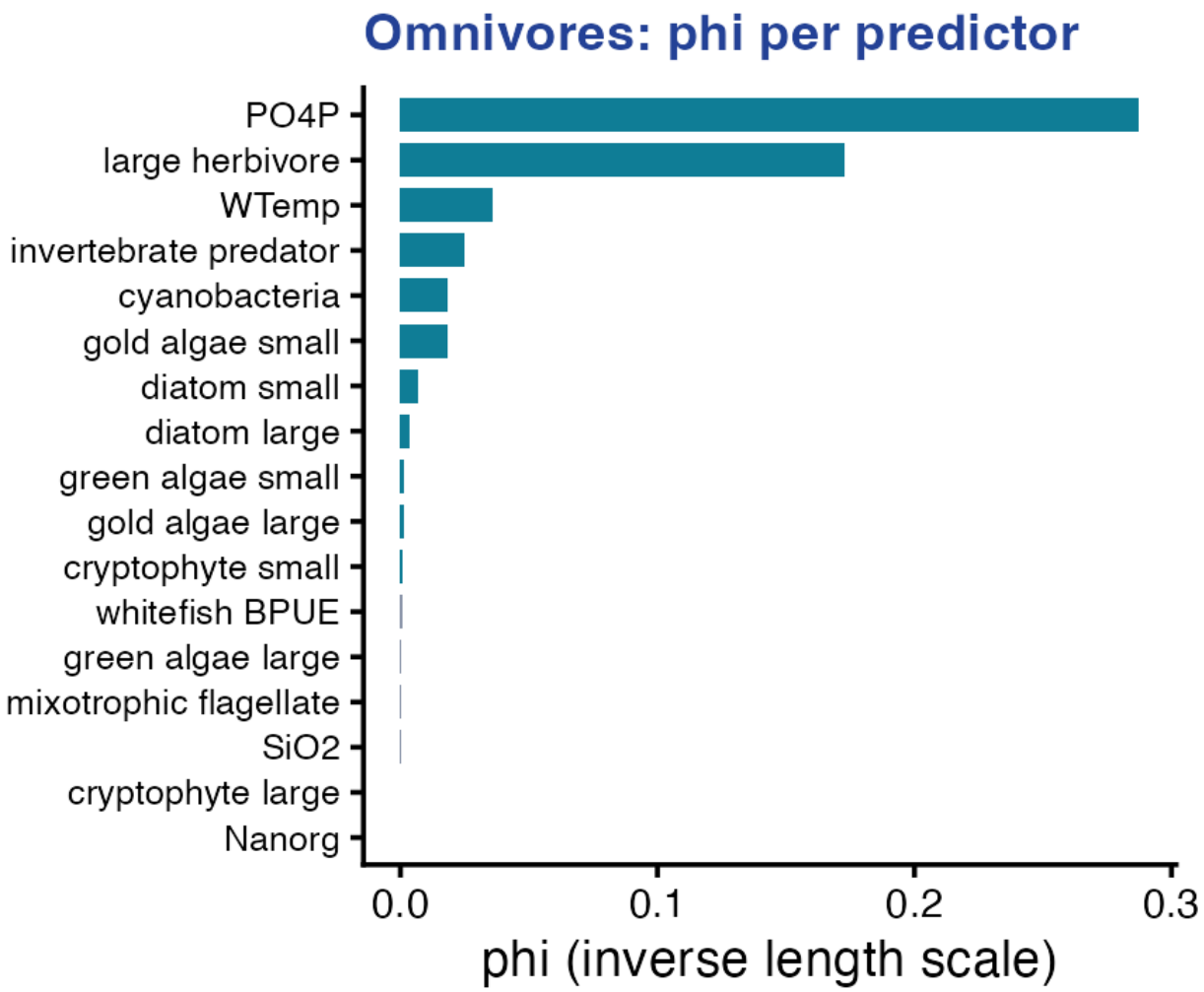
$\phi_2 = 0 \Rightarrow x_2$ never changes k

$\Rightarrow$ less noise, better LOO fit

Rule

$\phi_i \leq 0.001$: dropped

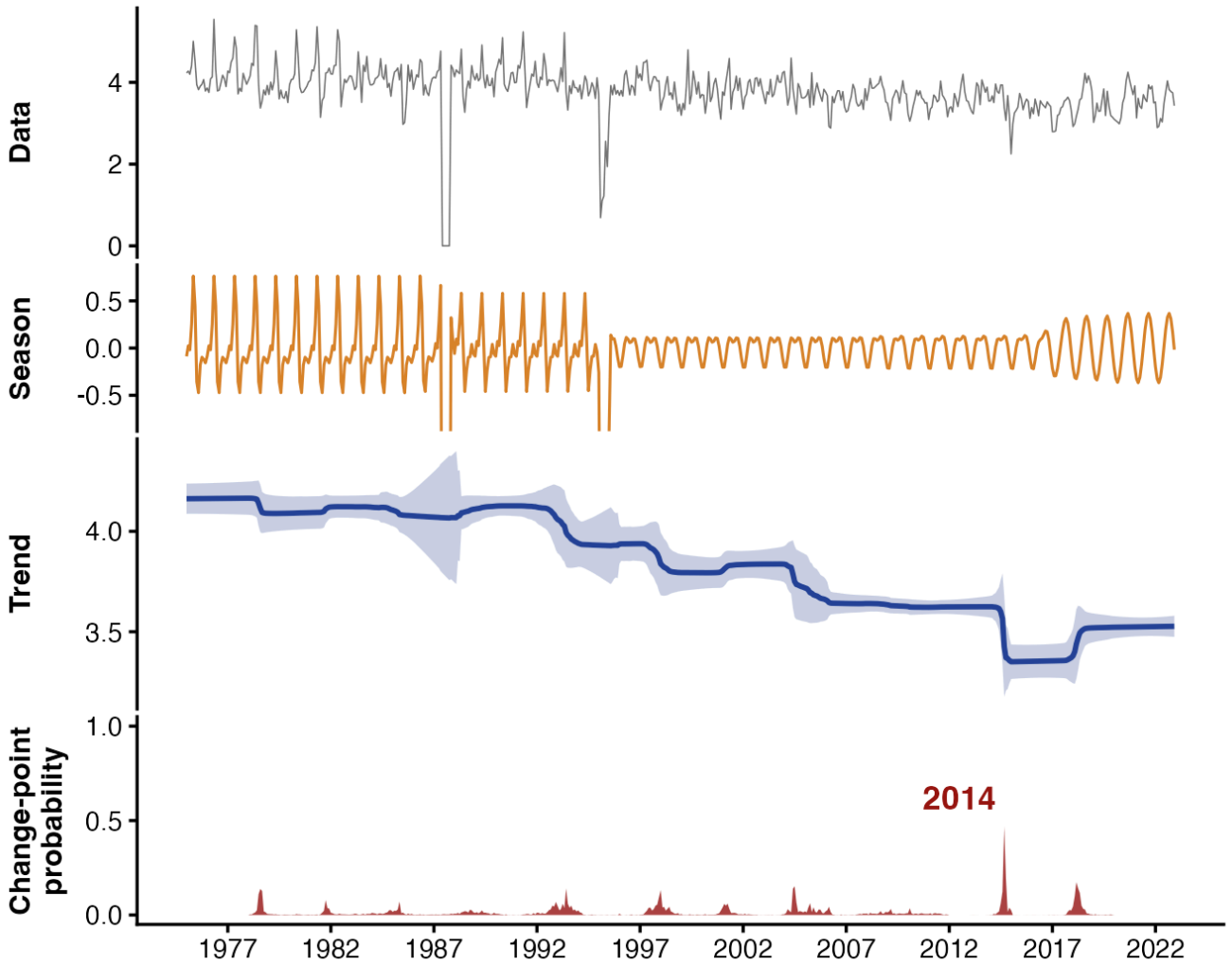
ARD in the omnivore model



Large ϕ
kept: phosphate, large herbivores, temperature ...

ϕ ≈ 0
dropped: nitrogen, silica, large cryptophytes ...

What does BEAST do?



Splits the series

y_t = trend + season + noise

Finds change points

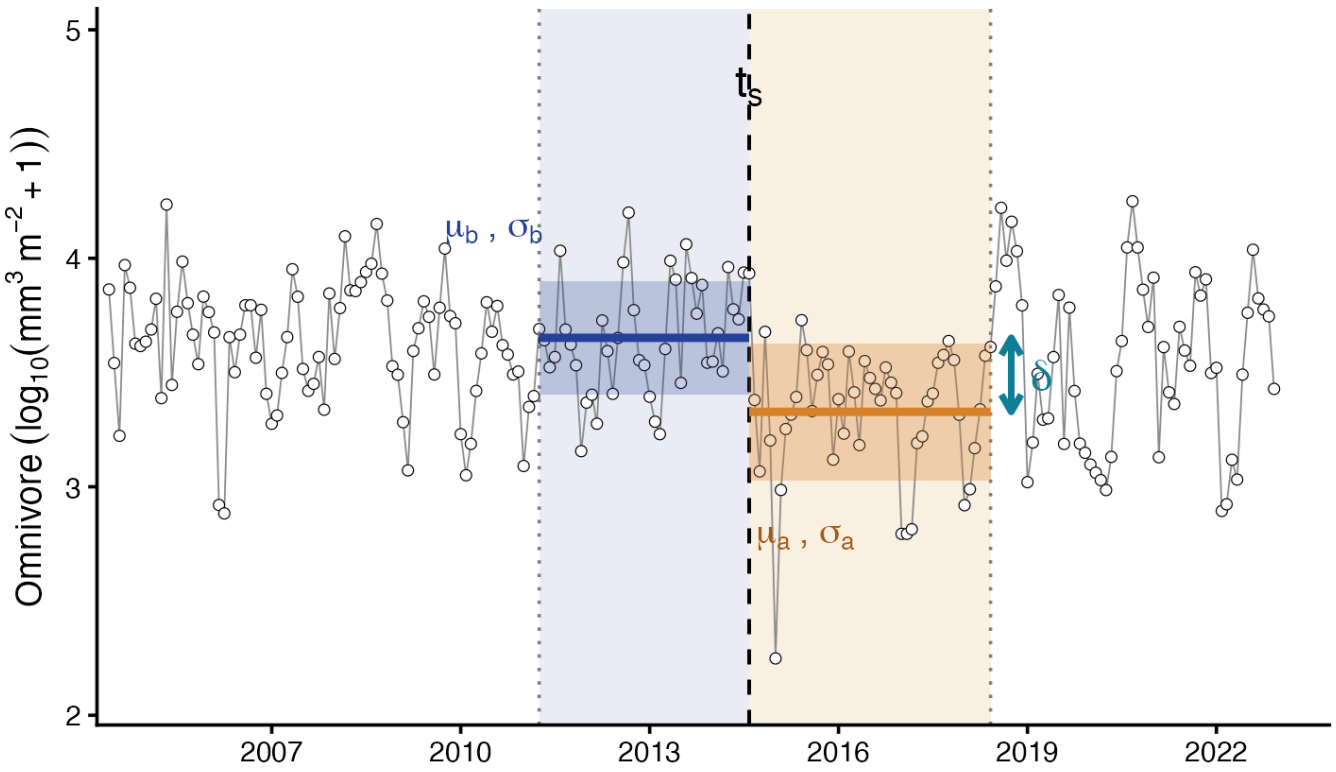
jumps of the **trend**

Gives a probability

Pr(change point) per month

Illustration: BEAST on the omnivores alone

Detecting regime shifts: criteria



1 Difference of the two chunks

$$\delta = \mu_a - \mu_b \quad 0 \notin CI_{95\%}(\delta)$$

CI = [0.19, 0.42] ✓

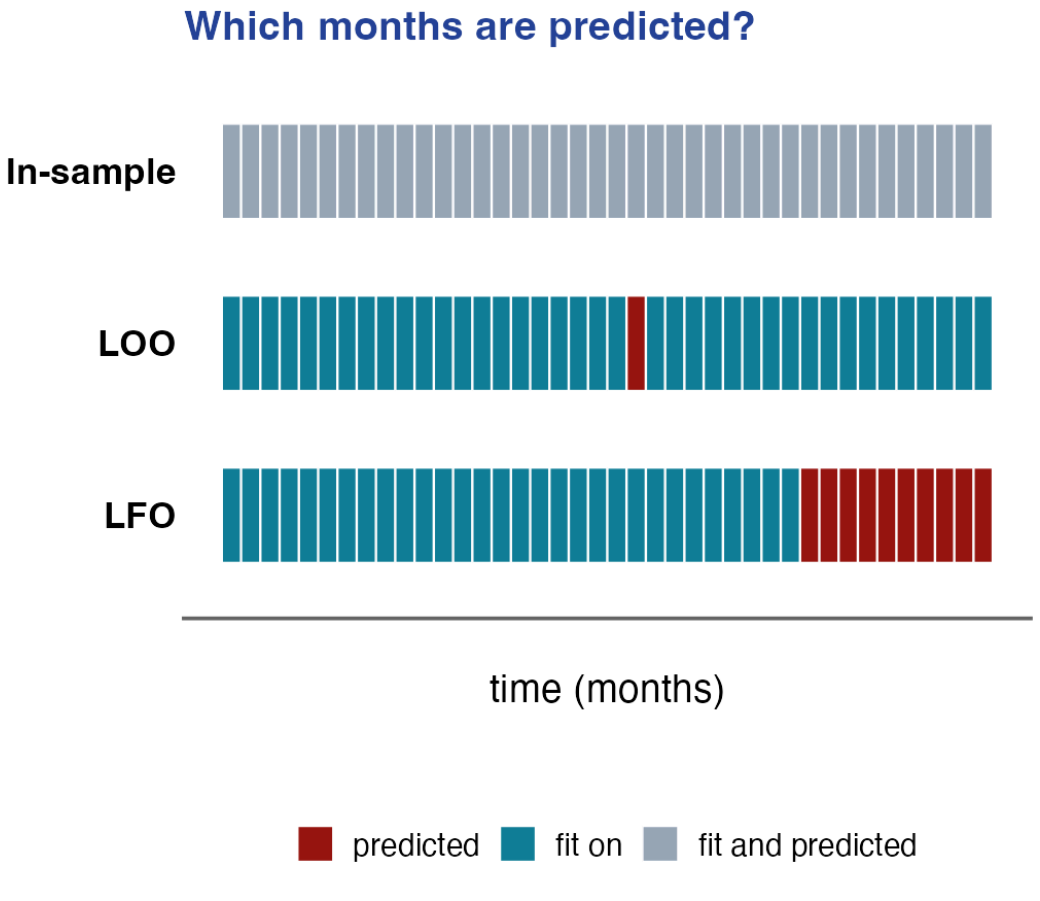
2 Abruptness: jump vs. usual swings

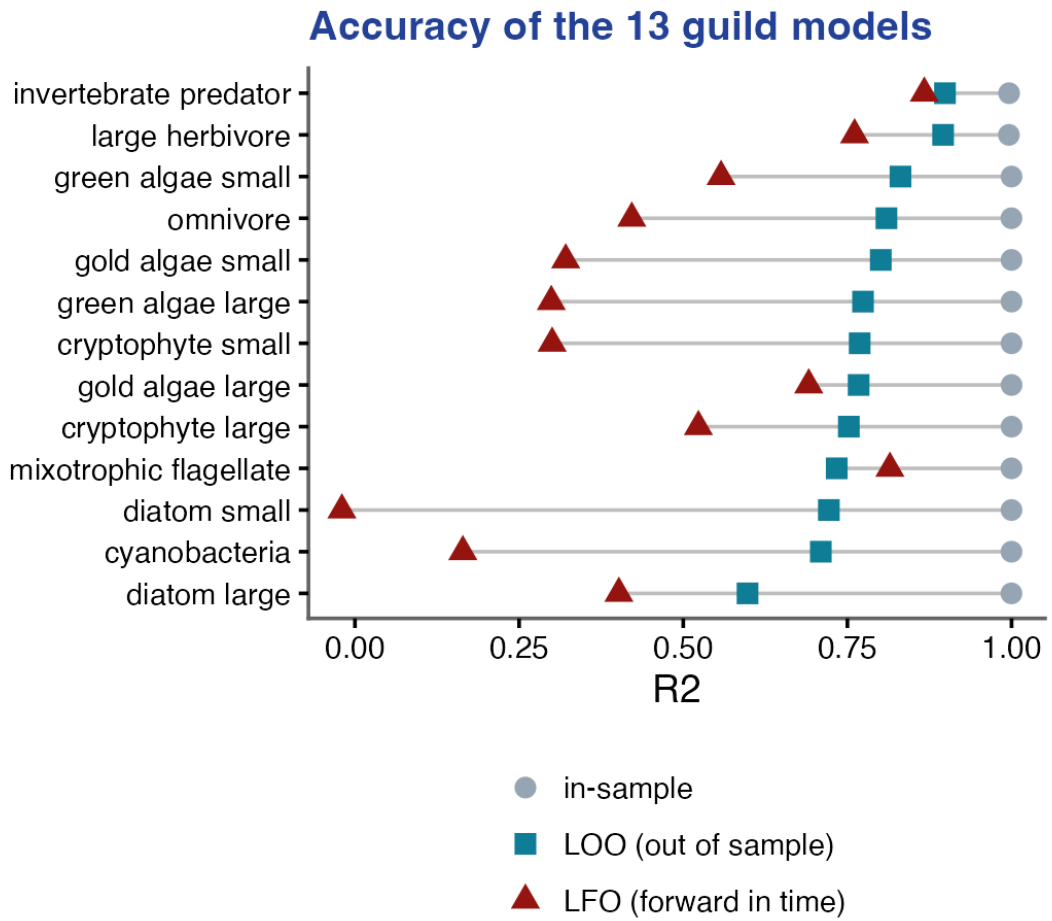
$$a = \frac{|\mu_a - \mu_b|}{½(\sigma_b + \sigma_a)} > 1$$

a = 1.19 ✓

Kept if 1 and 2 hold

In-sample versus out-of-sample R^2





In-sample: $R^2 = 1$, a GP interpolates

LOO: one month left out

LFO: forecast the last 10 %

Which predictors were used?

Omnivores LOO R^2 = 0.81

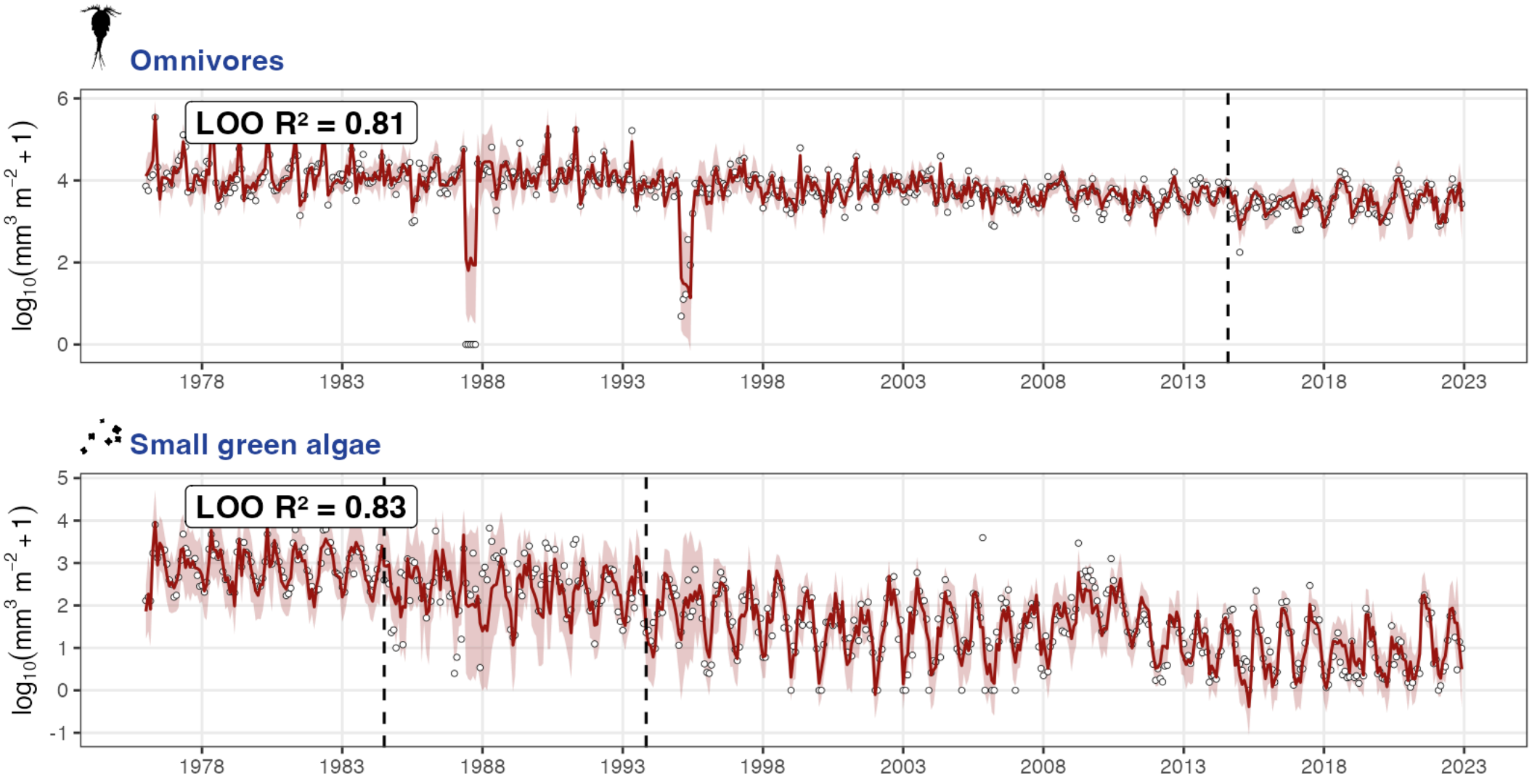
- Phosphate-P — 6 lags
- Large herbivores — 4 lags
- Whitefish — 4 lags
- Invertebrate predators — 7 lags
- Temperature — 4 lags
- Small green algae — 3 lags
- Small cryptophytes — 3 lags
- Large gold algae — 2 lags
- Small diatoms — 2 lags
- Cyanobacteria — 1 lags
- Large green algae — 2 lags
- Small gold algae — 2 lags
- Large diatoms — 2 lags
- Mixotrophic flagellates — 1 lags
- Silica — 1 lags
- Large cryptophytes — removed by ARD
- Inorganic N — removed by ARD

Small green algae LOO R^2 = 0.83

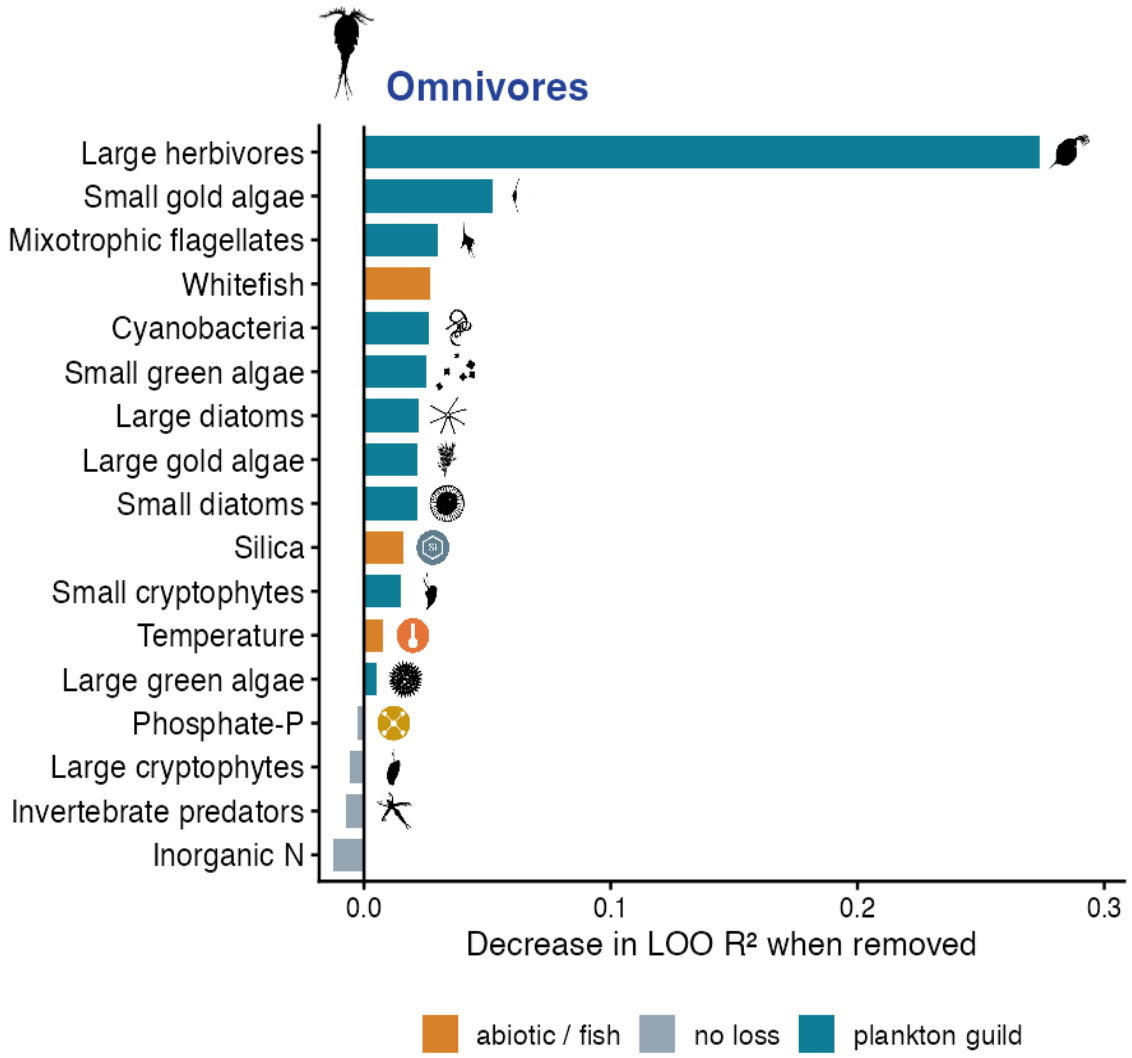
- Phosphate-P — 3 lags
- Large green algae — 8 lags
- Large cryptophytes — 4 lags
- Temperature — 2 lags
- Large gold algae — 4 lags
- Omnivores — 6 lags
- Large herbivores — 4 lags
- Mixotrophic flagellates — 3 lags
- Invertebrate predators — 3 lags
- Large diatoms — 3 lags
- Cyanobacteria — 4 lags
- Whitefish — 5 lags
- Small cryptophytes — 3 lags
- Small gold algae — 3 lags
- Small diatoms — 3 lags
- Inorganic N — not entered
- Silica — not entered

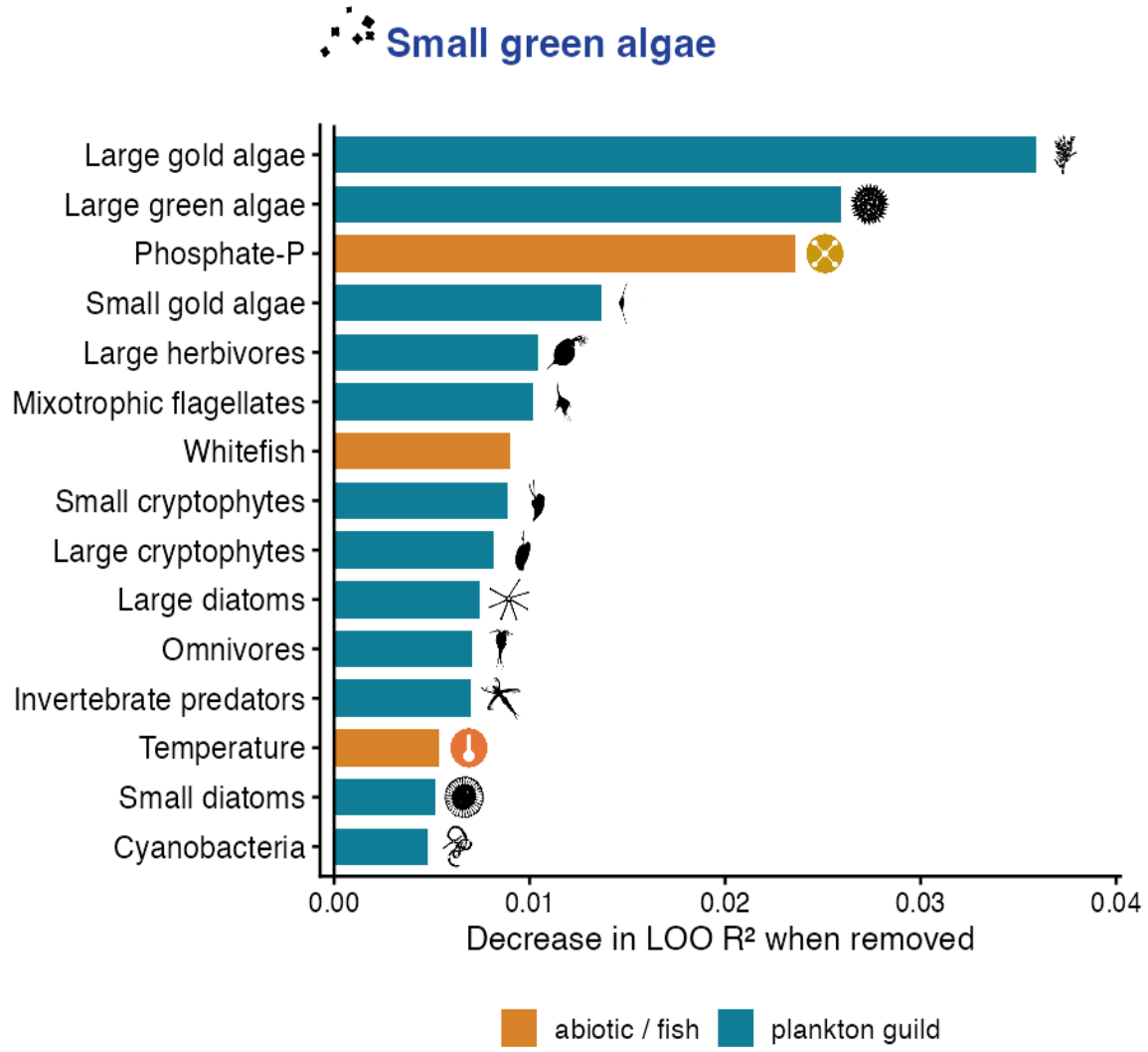
Candidate predictors: all other guilds, temperature, phosphate, silica, inorganic N and whitefish, each at lags 0–12 months

Model fit: LOO R^2



Which predictors matter?





Which genera? Zooplankton and mixotrophs



Invertebrate predators

Pr

Leptodora	73%
Bythotrephes	27%



Omnivores

Om

Cyclops	80%
Mesocyclops	17%
Acanthocyclops	1%
Thermocyclops	1%
Asplanchna	0%

+ 4 more



Large herbivores

He

Daphnia	91%
Eudiaptomus	6%
Bosmina	2%
Diaphanosoma	1%
Acanthodiaptomus	0%

+ 6 more



Mixotrophic flagellates

Mi

Ceratium	46%
Gymnodinium	45%
Peridinium	7%
other Flagellaten	1%
other Phytoflagellaten	0%

+ 14 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (1/2)



Cyanobacteria

Cy colonial

Anabaena	44%
Aphanizomenon	16%
Planktothrix	9%
Cyanothece	7%
Aphanothece	6%

+ 25 more



Green algae, small

Gr1 ≤ 550 µm³

Volvocales	30%
unidentified algae	24%
Chlorella	21%
Oocystis	7%
Characium	3%

+ 21 more

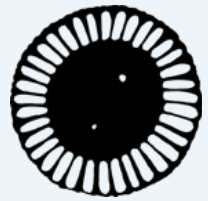


Green algae, large

Gr2 > 550 µm³ or colonial

other Chlorococcales	23%
Pandorina	18%
Mougeotia	16%
Chlamydomonas	10%
Ulothrix	6%

+ 46 more

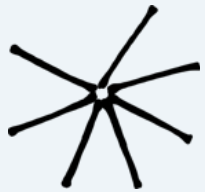


Diatoms, small

Di1 ≤ 800 µm³

Stephanodiscus	81%
Cyclotella	17%
Achnanthes	1%
Fragilaria	1%
Nitzschia	0%

+ 9 more



Diatoms, large

Di2 > 800 µm³ or colonial

Fragilaria	50%
Stephanodiscus	15%
Asterionella	14%
Cyclotella	10%
Tabellaria	4%

+ 18 more

Top genera by share of biovolume, IGKB 1975–2022

Which genera? Phytoplankton (2/2)



Gold algae, small

Go1 ≤ 250 µm³

Erkenia	63%
Chrysochromulina	23%
Chromulina	4%
Ochromonas	2%
Pseudokephyrion	2%

+ 13 more



Gold algae, large

Go2 > 250 µm³ or colonial

Dinobryon	42%
Mallomonas	17%
Uroglena	12%
Chrysophyceae	12%
Pseudopedinella	6%

+ 9 more



Cryptophytes, small

Cr1 ≤ 700 µm³

Rhodomonas	99%
Plagioselmis	1%
Cyathomonas	0%
Cryptaulax	0%
Chilomonas	0%

+ 1 more



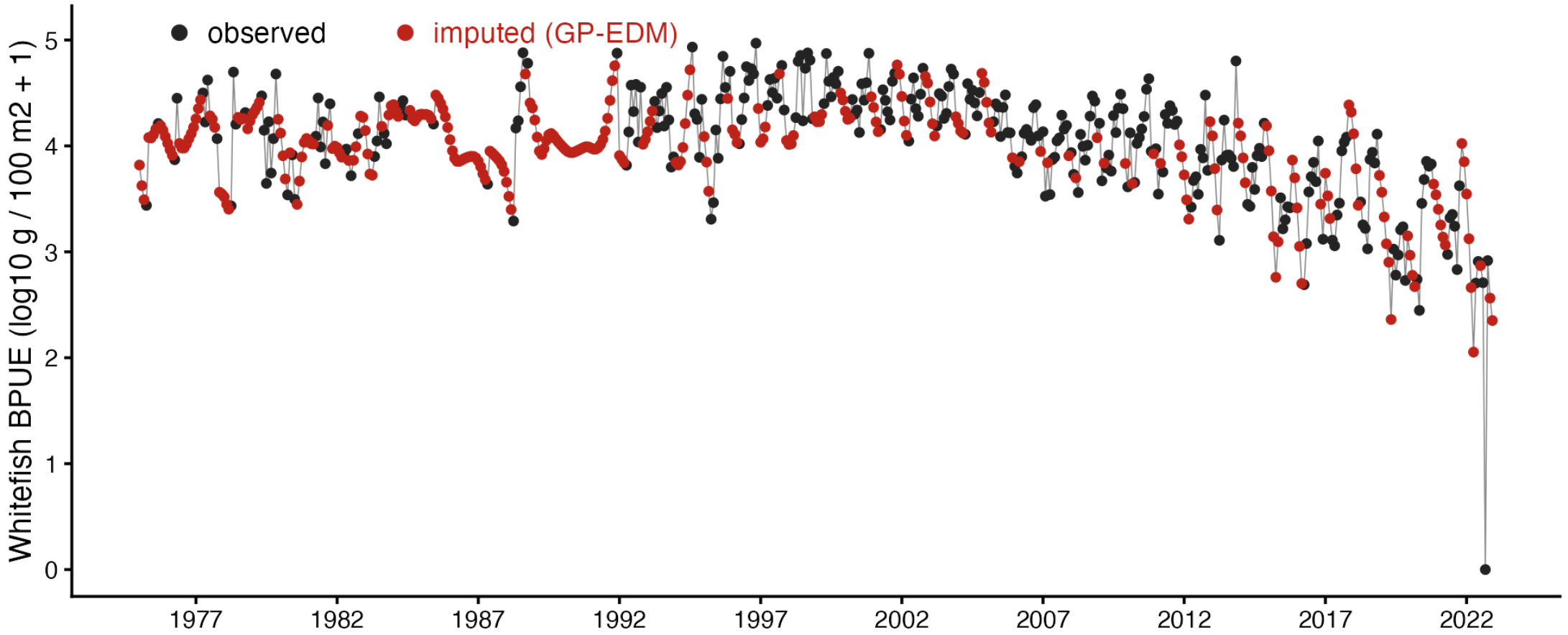
Cryptophytes, large

Cr2 > 700 µm³

Cryptomonas	100%
other Cryptophyceen	0%

Top genera by share of biovolume, IGKB 1975–2022

Whitefish time series: observed and imputed



287 of 576 months imputed (50 %)

GP-EDM, LOO R^2 = 0.55

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Lake Constance map (panel a): adapted from the published figure provided by the author. Panels b–d: IGKB monitoring data (phosphate, temperature) and IBKF whitefish catches, annual means 1975–2022.